

3: Introduction to probability

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Probability

How often, on average, does an event occur?

- Probability is a set of tools for describing randomness.
- Probability helps us sort signal (patterns) from noise.

Two core theories

Frequentist: Probability is the proportion of times an event occurs if we repeat an experiment under the same conditions many times

Consider a simple experiment

- What is the probability of obtaining a heads on a single fair coin flip?

```
1 flip_n_coins<-function(x){  
2   flip<-rbinom(x, 1, 0.5)  
3   flip<-ifelse(flip==1, "Heads", "Tails")  
4   return(flip)  
5 }
```

```
1 flip_n_coins(1)
```

```
[1] "Heads"
```

```
1 flip_n_coins(1)
```

```
[1] "Heads"
```

```
1 flip_n_coins(1)
```

```
[1] "Tails"
```

What is $\frac{\sum_1^n x_i}{n}$ where $x_i = 1$ when the coin is heads?

Probability as the average outcome under repeat experiments

```
1 sum((flip_n_coins(5)=="Heads")/5)
```

```
[1] 0.8
```

```
1 sum((flip_n_coins(20)=="Heads")/20)
```

```
[1] 0.4
```

```
1 sum((flip_n_coins(50)=="Heads")/50)
```

```
[1] 0.56
```

```
1 sum((flip_n_coins(1000)=="Heads")/1000)
```

```
[1] 0.473
```

```
1 sum((flip_n_coins(100000)=="Heads")/100000)
```

[1] 0.50121

Frequentist probability

If n_x is the number of heads, and n_t is the number of coin flips, then the probability of heads is

$$P(x) \approx \frac{n_x}{n_t}$$

$$P(x) = \lim_{n_t \rightarrow \infty} \frac{n_x}{n_t}$$

Bayesian probability

Bayesian: Probability is a subjective judgment about the likelihood that an event occurs, with endpoints at 0 (never occurs) and 1 (always occurs).

Definitions

Deterministic processes do not include randomness. For example, if I drop a ball, it will fall (because gravity).

Stochastic events include randomness. For example, if I flip a fair coin, it will be heads half of the time.

Nearly all social processes are have random components, and can be treated as *stochastic*.

Definitions and axioms

- Experiment: an action that produces stochastic events
- Sample space (Ω): a set of all possible outcomes of the experiment
- Event: a subset of the sample space

Example: coin flips

- Experiment
 - . flip a coin
 - . roll a dice
- Sample space Ω
 - . {Heads, Tails}
 - . {1,2,3,4,5,6}
- Event
 - . Heads, tails, not heads, heads or tails, heads and tails
 - . 3, even number, anything but 6

Probability axioms

- . The probability of any event A is non-negative: $P(A) \geq 0$
- . If an experiment is conducted, the probability that one of the outcomes in the sample space occurs is 1: $P(\Omega) = 1$
- . Addition rule: If events A and B are mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$

Probability that an event doesn't occur

$$1 - P(\text{not } A) = P(A)$$

$$\text{If } P(\text{rolling a 6}) = \frac{1}{6} \text{ then } P(\text{not rolling a 6}) = 1 - \frac{1}{6} = \frac{5}{6}$$

Law of total probability

$$P(A) = P(A \text{ and } B) + P(A \text{ and not } B)$$

Assume the sample space for A is [eats pizza, does not eat pizza], and the sample space for B is [happy, unhappy].

If $P(\text{eats pizza}) = 0.5$ and $P(\text{eats pizza, happy}) = 0.4$, then

$$P(\text{eats pizza, unhappy}) = 0.1$$

General addition rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

If $P(\text{happy}) = 0.5$, then $P(\text{eats pizza or happy}) = 0.6$

Three kinds of probability

Joint probability

The joint probability of two events (A and B) occurring is expressed as $P(A \text{ and } B)$

Marginal probability

The marginal probability of an event B is

$$P(B)$$

Conditional probability

The conditional probability of event A occurring given that event B occurred is

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

Working with some real data

Voter files

```
1 data("FLVoters")
2 voters<-na.omit(FLVoters)
3 head(voters)
```

	surname	county	VTD	age	gender	race
1	PIEDRA	115	66	58	f	white
2	LYNCH	115	13	51	m	white
4	LATHROP	115	80	54	m	white
5	HUMMEL	115	8	77	f	white
6	CHRISTISON	115	55	49	m	white
7	HOMAN	115	84	77	f	white

Marginal probability

What is the probability that a randomly sampled voter in the population is Black:
 $P(\textit{Black}) = ?$

```
1 voters |>  
2   count(race) |>  
3   mutate(p = n/sum(n))
```

	race	n	p
1	asian	175	0.019203336
2	black	1194	0.131021617
3	hispanic	1192	0.130802151
4	native	29	0.003182267
5	other	310	0.034017338
6	white	6213	0.681773291

Is a woman: $P(\textit{Woman}) = ?$

```
1 voters |>  
2   count(gender) |>  
3   mutate(p = n/sum(n))
```

	gender	n	p
1	f	4883	0.5358279
2	m	4230	0.4641721

Joint probability

What is the probability that a voter is a Black woman: $P(\text{Black and woman}) = ?$

```
1 voters |>  
2   count(gender, race) |>  
3   mutate(n = n/sum(n)) |>  
4   pivot_wider(names_from = gender, values_from = n)
```

```
# A tibble: 6 × 3  
  race      f      m  
  <chr>  <dbl> <dbl>  
1 asian  0.00911 0.0101  
2 black  0.0744  0.0566  
3 hispanic 0.0731 0.0577  
4 native 0.00187 0.00132  
5 other  0.0173  0.0167  
6 white  0.360   0.322
```

What is the probability that a voter is a woman?

```
# A tibble: 6 × 3
  race      f      m
  <chr>    <dbl>  <dbl>
1 asian  0.00911 0.0101
2 black  0.0744  0.0566
3 hispanic 0.0731 0.0577
4 native 0.00187 0.00132
5 other  0.0173  0.0167
6 white  0.360   0.322
```

Use the law of total probability:

$$P(A) = P(A \text{ and } B) + P(A \text{ and not } B)$$

put differently, for all categories of B i :

$$P(A) = \sum_{i=1}^n P(A \text{ and } B_i)$$

Conditional probability

If a voter is a man, what is the probability that he is Asian: $P(\text{Asian}|\text{man}) = ?$

```
1 voters |>  
2   filter(gender=="m") |>  
3   count(race) |>  
4   mutate(p=n/sum(n))
```

	race	n	p
1	asian	92	0.021749409
2	black	516	0.121985816
3	hispanic	526	0.124349882
4	native	12	0.002836879
5	other	152	0.035933806
6	white	2932	0.693144208

Conditioning on more than one variable

What is the probability that a male voter over age 60 is white?

$P(\text{whitemale and over 60})$

```
1 voters |>
2   mutate(over60=age>60, white = race=="white") |>
3   count(over60, gender, white) |>
4   mutate(n=n/sum(n)) |>
5   pivot_wider(names_from = gender, values_from = n)
```

A tibble: 4 × 4

	over60	white	f	m
	<lgl>	<lgl>	<dbl>	<dbl>
1	FALSE	FALSE	0.131	0.109
2	FALSE	TRUE	0.212	0.198
3	TRUE	FALSE	0.0449	0.0335
4	TRUE	TRUE	0.148	0.124

Independence

Two events are independent if knowledge of one event gives us no information about the other event.

$$P(A|B) = P(A) \text{ and } P(B|A) = P(B)$$

$$A \perp B$$

if and only if

$$P(A \text{ and } B) = P(A)P(B)$$

Bayes' rule

How should we change our beliefs after we make observations about the world?

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Prior beliefs and evidence

If we have a *prior* belief that event A has $P(A)$ chance of occurring, we observe some data, B , we update our beliefs and obtain a *posterior probability* $P(A|B)$.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Example: Detecting breast cancer

How good is a mammogram at detecting breast cancer?

One percent of women have breast cancer. 80 percent of people who have cancer and take a mammogram test positive. 9.6 percent of mammograms are a false positive

If you take a mammogram and get a positive result, what is the probability that you have breast cancer?

Rewriting as probabilities

One percent of women have breast cancer

$$P(\text{Cancer}) = 0.01$$

80 percent of people who have cancer and take a mammogram test positive

$$P(\text{Test positive}|\text{Cancer}) = 0.8$$

9.6 percent of people who take a mammogram get a positive result when they do not have breast cancer

$$P(\text{Test positive}|\text{No cancer}) = 0.096$$

Using Bayes' rule

The prior probability of having cancer is 0.01. How should we update our belief that someone has cancer based on a positive test?

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Using the law of total probability, we can rewrite the denominator as:

$$P(B) = P(B|A)P(A) + P(B|\text{not } A)P(\text{not } A)$$

Using Bayes' rule

We can apply Bayes' rule for A = Cancer, B = positive test:

$$P(\text{Cancer}|\text{Test positive}) = \frac{P(\text{Test positive}|\text{Cancer})P(\text{Cancer})}{P(\text{Test positive})}$$

$$P(\text{Cancer}|\text{Test positive}) = \frac{0.8 \times 0.01}{0.8 \times 0.01 + 0.096 \times 0.99}$$

```
1 (0.8 * 0.01)/(0.8 * 0.01 + 0.096 * 0.99)
```

```
[1] 0.07763975
```

Random Variables

Random Variables

- Random variables assign values to events
- Each value is mutually exclusive
- The set of all values is exhaustive (the sample space Ω)
- Discrete random variables take a finite number of values (e.g. TRUE, FALSE)
- Continuous random variables are real numbers, and take on an infinite number of values

The simplest random variable: the binary Bernoulli

- Any random variable with two values is called a Bernoulli random variable.
- Bernoulli (binary) variables are typically represented as $[0, 1]$ or $[T, F]$.

A coin flip as a Bernoulli random variable

A coin flip can be defined as a discrete random variable X

- If the coin lands on heads, $X = 1$
- If the coin lands on tails, $X = 0$
- The probability of a Bernoulli variable is the probability of success, or $X = 1$
- $P(X = 1) = p$

Random variable (probability distribution) notation

$$X \sim \textit{Bernoulli}(p)$$

Reads: X is a Bernoulli distributed random variable with probability p

Probability mass

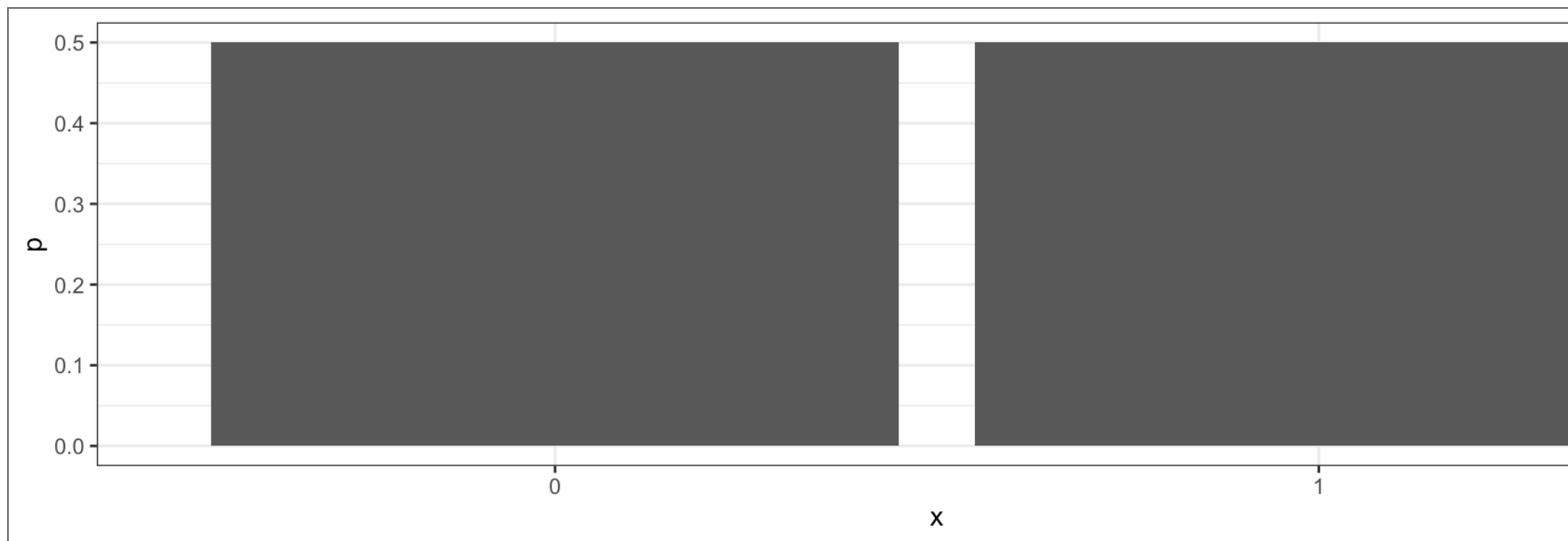
The probability mass function (PMF) of a variable X is defined as the probability that a variable takes on a particular value x .

$$PMF(x) = P(X = x)$$

PMF for a coin flip

$$PMF(X = 1) = p = 0.5$$

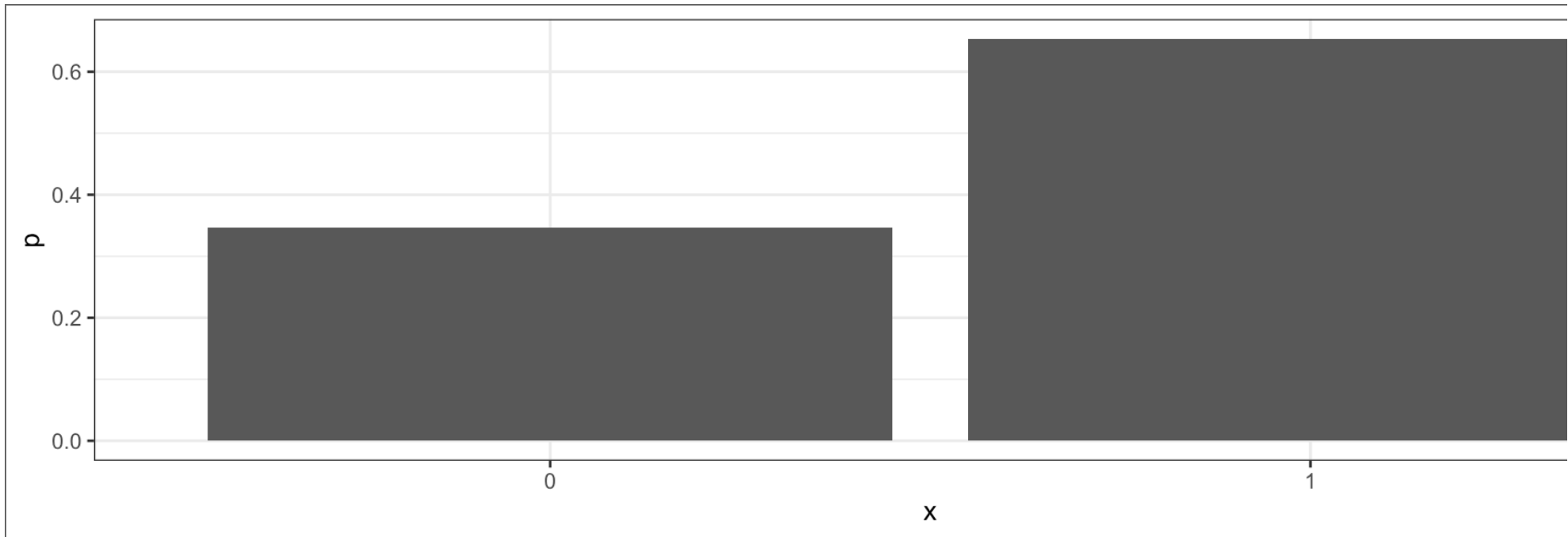
$$PMF(X = 0) = 1 - p = 0.5$$



The probability mass function for passing the bar in NJ ($p=0.653$)

$$PMF(X = 1) = p = 0.653$$

$$PMF(X = 0) = 1 - p = 0.347$$



Describing a probability distribution: cumulative probability

- How likely is a variable to take a value less than or equal to a specified value?
- We define the cumulative distribution function as the sum of all probabilities up to a value x
- $$CDF(X) = P(X \leq x) = \sum_{k \leq x}^x PMF(k)$$
- The CDF always ranges from 0 to 1, and never decreases as x increases.

Uniform random variables

- Uniform random variables have an equal probability of taking any real value within a given interval $[a, b]$.
- What does $X \sim \text{Uniform}(0, 10)$ look like?

Uniform random variables

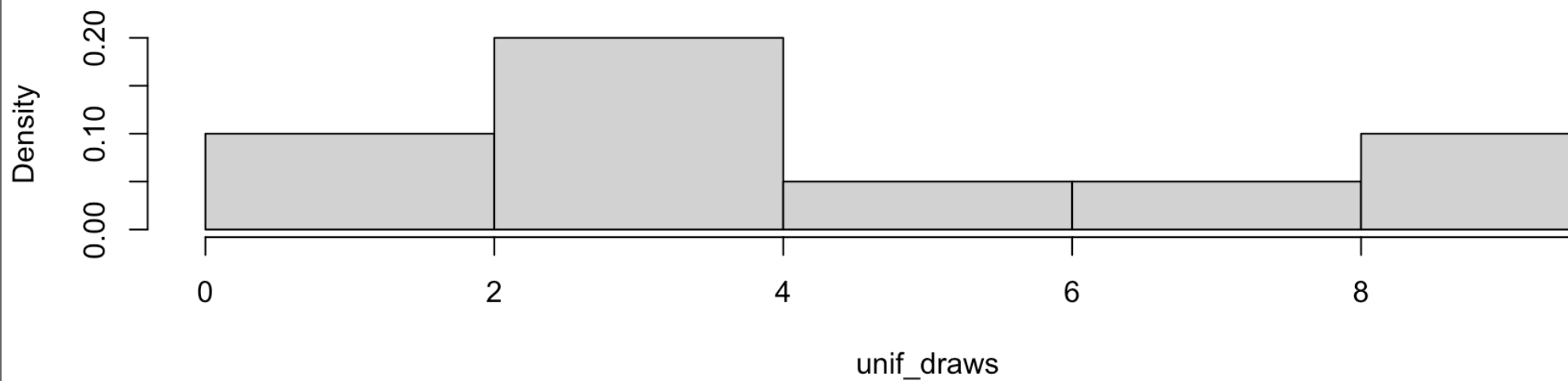
Let's simulate it! 10 draws

```
1 unif_draws<-runif(10, min=0, max=10)
2 unif_draws
```

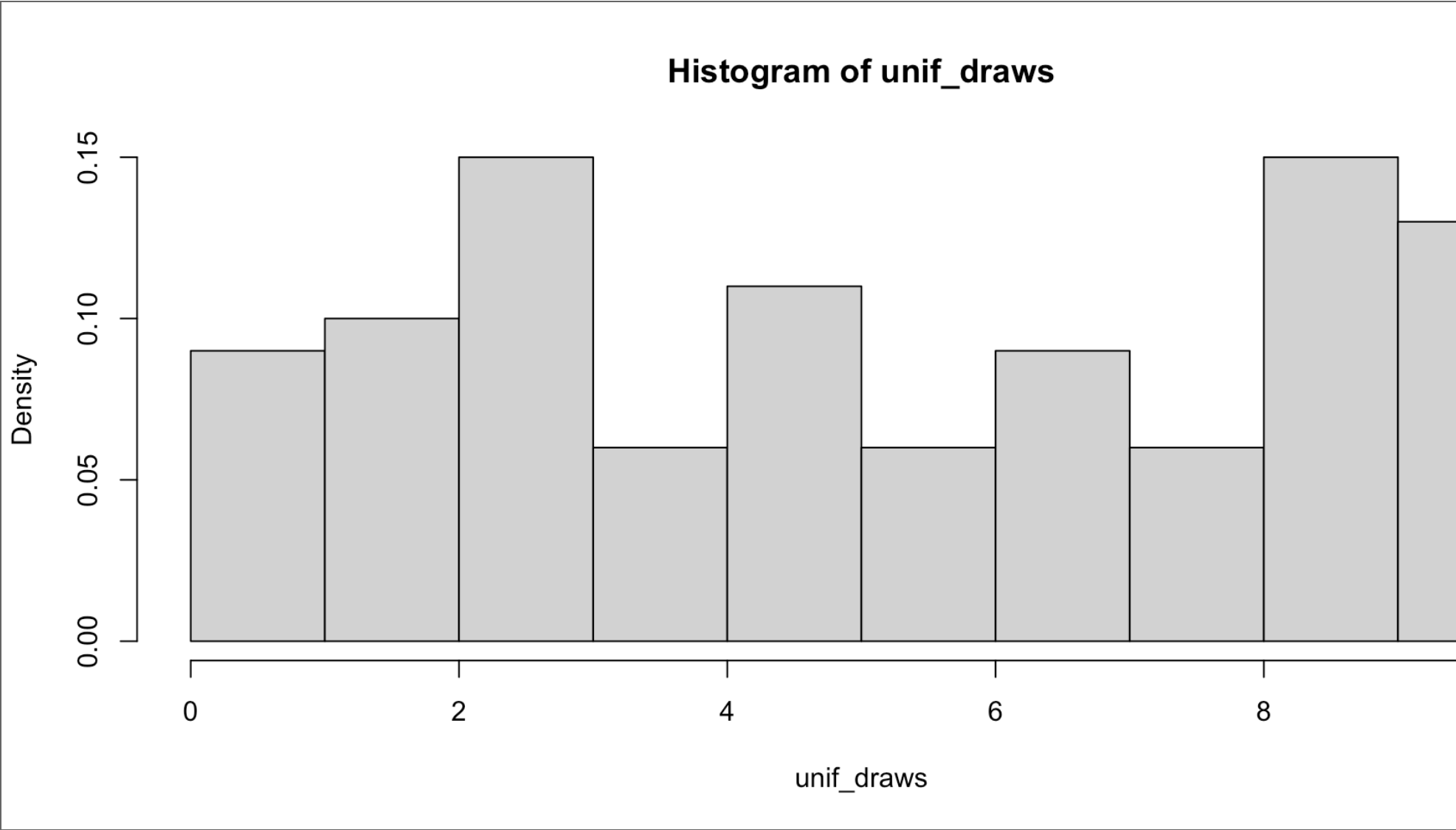
```
[1] 2.090964 5.376292 3.993732 9.425765 9.704637 1.260444 3.612929 1.832670
[9] 7.798734 2.021644
```

```
1 hist(unif_draws, freq=F)
```

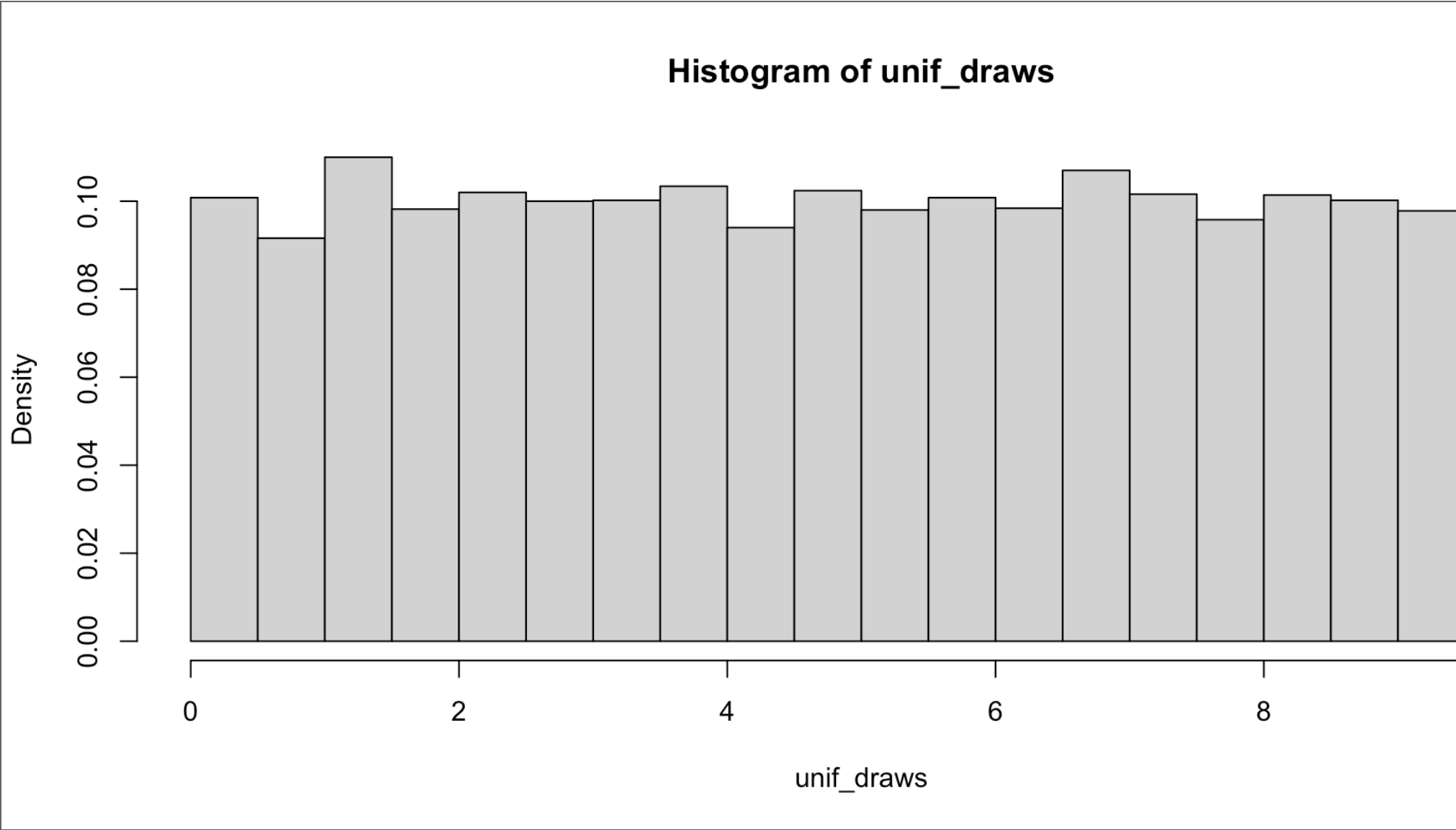
Histogram of unif_draws



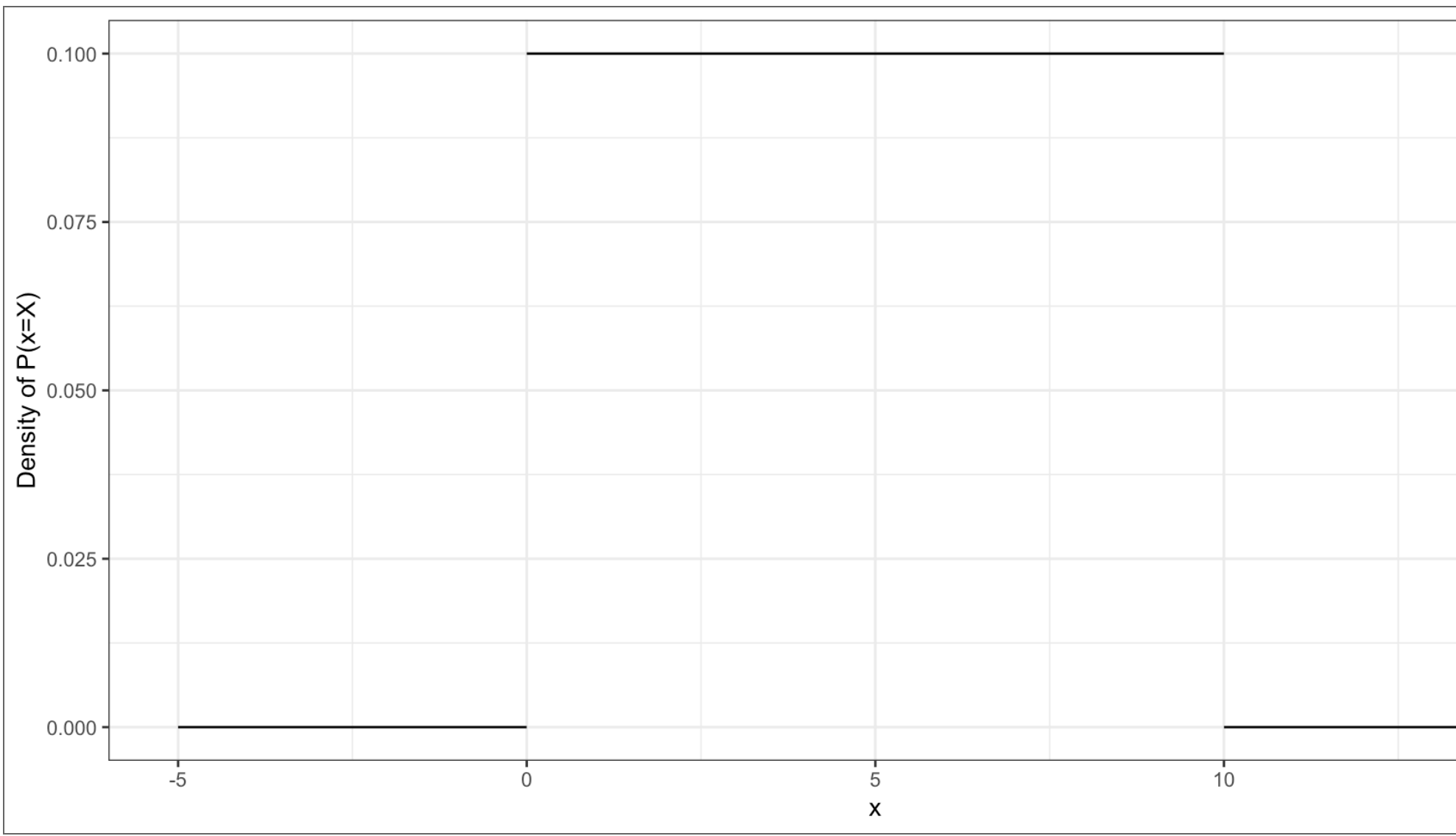
Uniform random variable: 100 draws



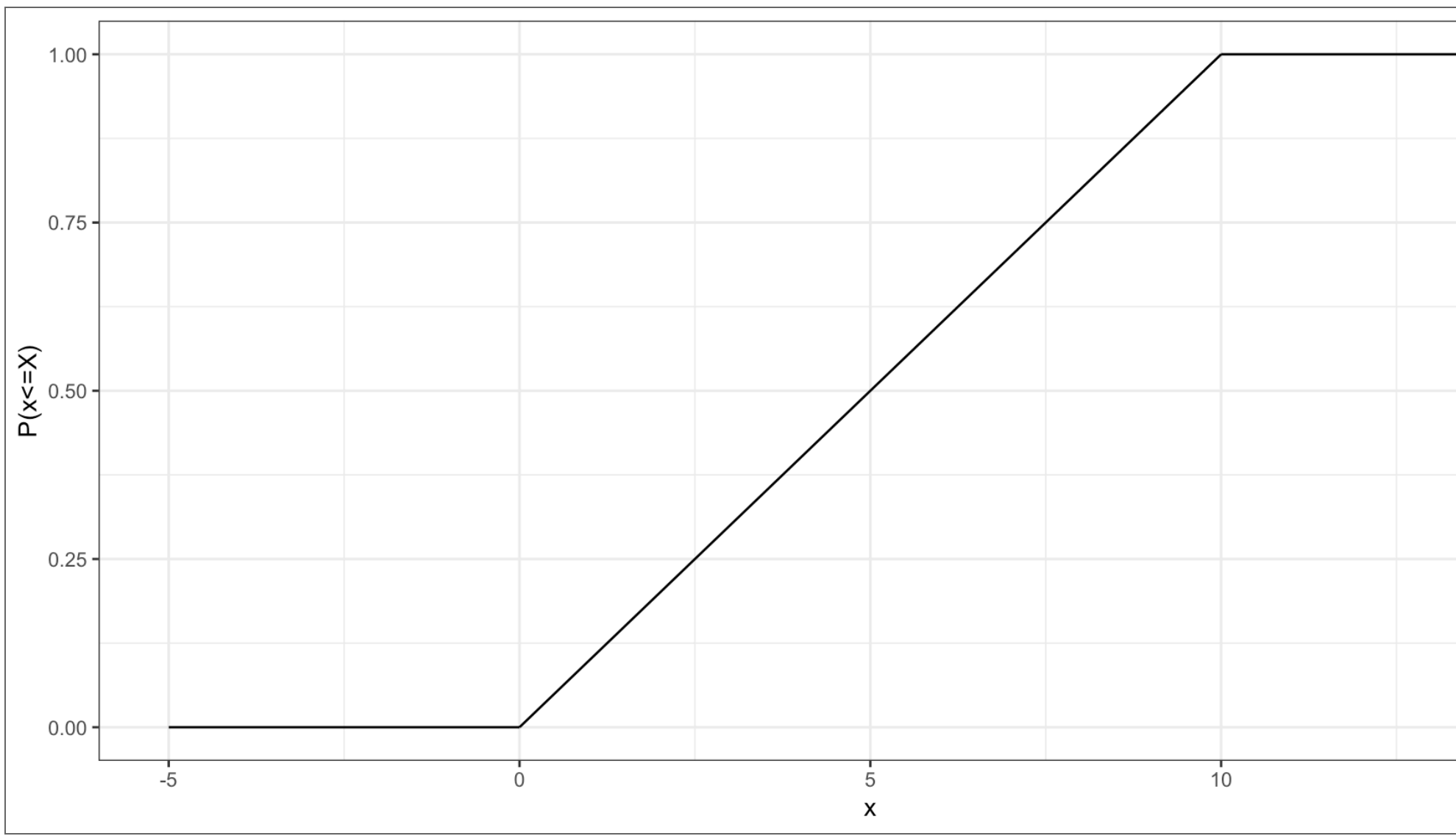
Uniform random variable: 10000 draws



PDF for $X \sim \text{Uniform}(0, 10)$



CDF for $X \sim \text{Uniform}(0, 10)$



The expectation of a random variable

- The expectation of a random variable $E(X)$ is the mean of a random variable.
- For a discrete variable, the expectation is the sum of all values of x weighted by their probability, given by the PDF $f(x)$.
- $$E(X) = \sum_x x \times f(x)$$

Variance and standard deviation

- The standard deviation sd is

- $$sd = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

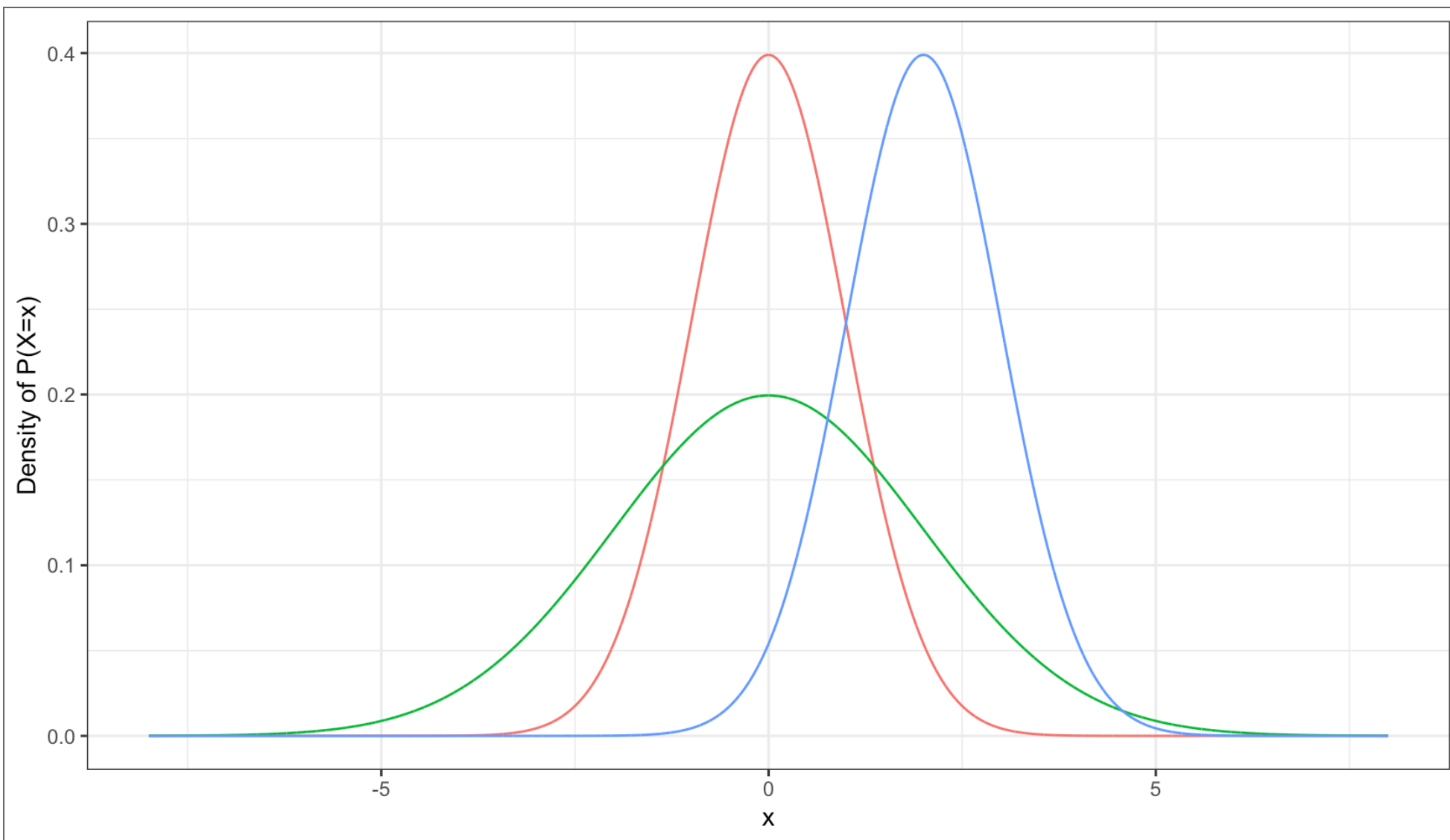
- And the sample variance is sd^2

The Normal Distribution

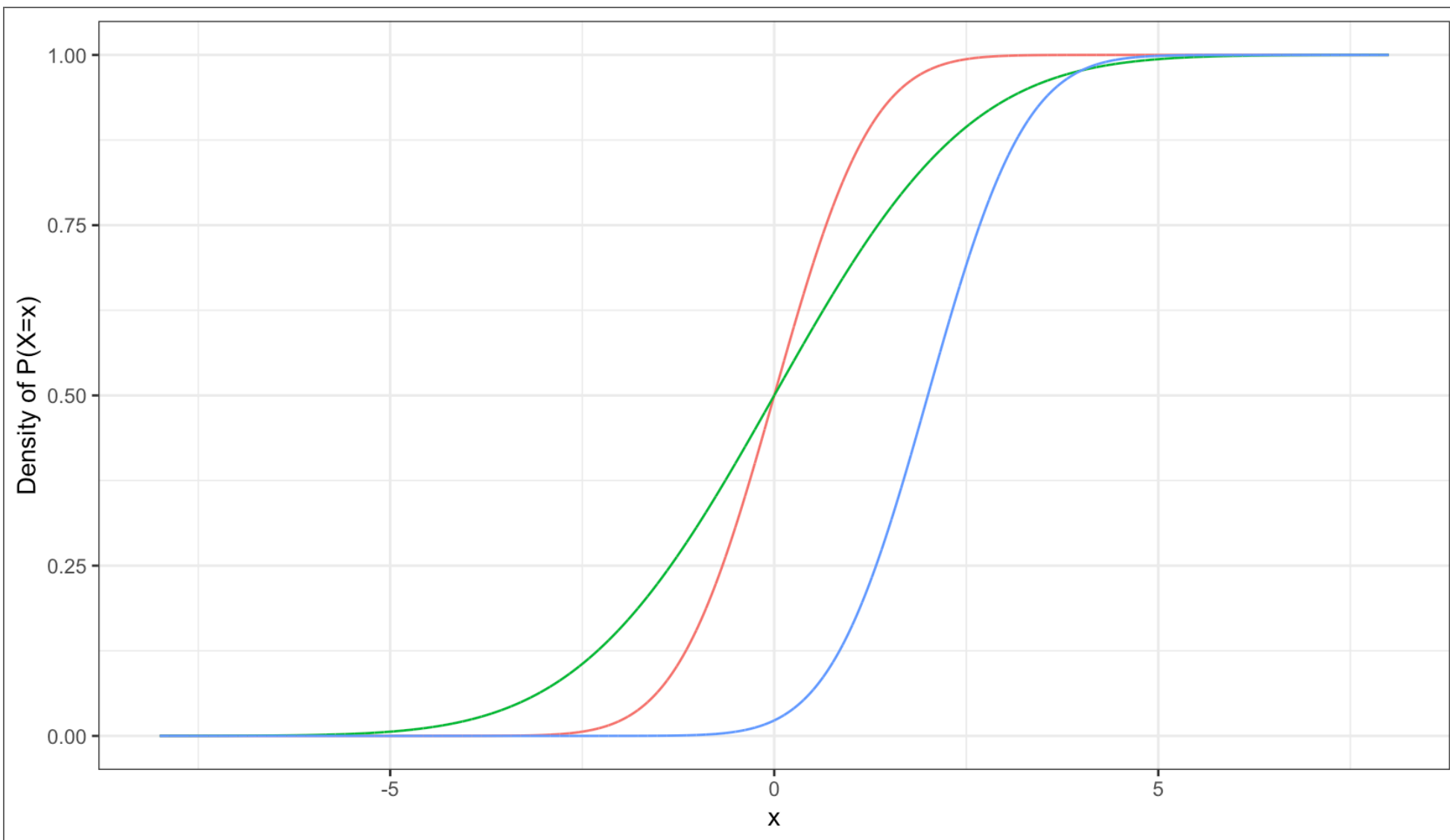
The Normal (Gaussian) distribution is continuous, and takes on values from $[-\infty, \infty]$. It has two parameters, the mean μ and standard deviation σ (or variance σ^2).

- μ determines the location of the distribution
- σ determines the spread of the distribution

The Normal PDF



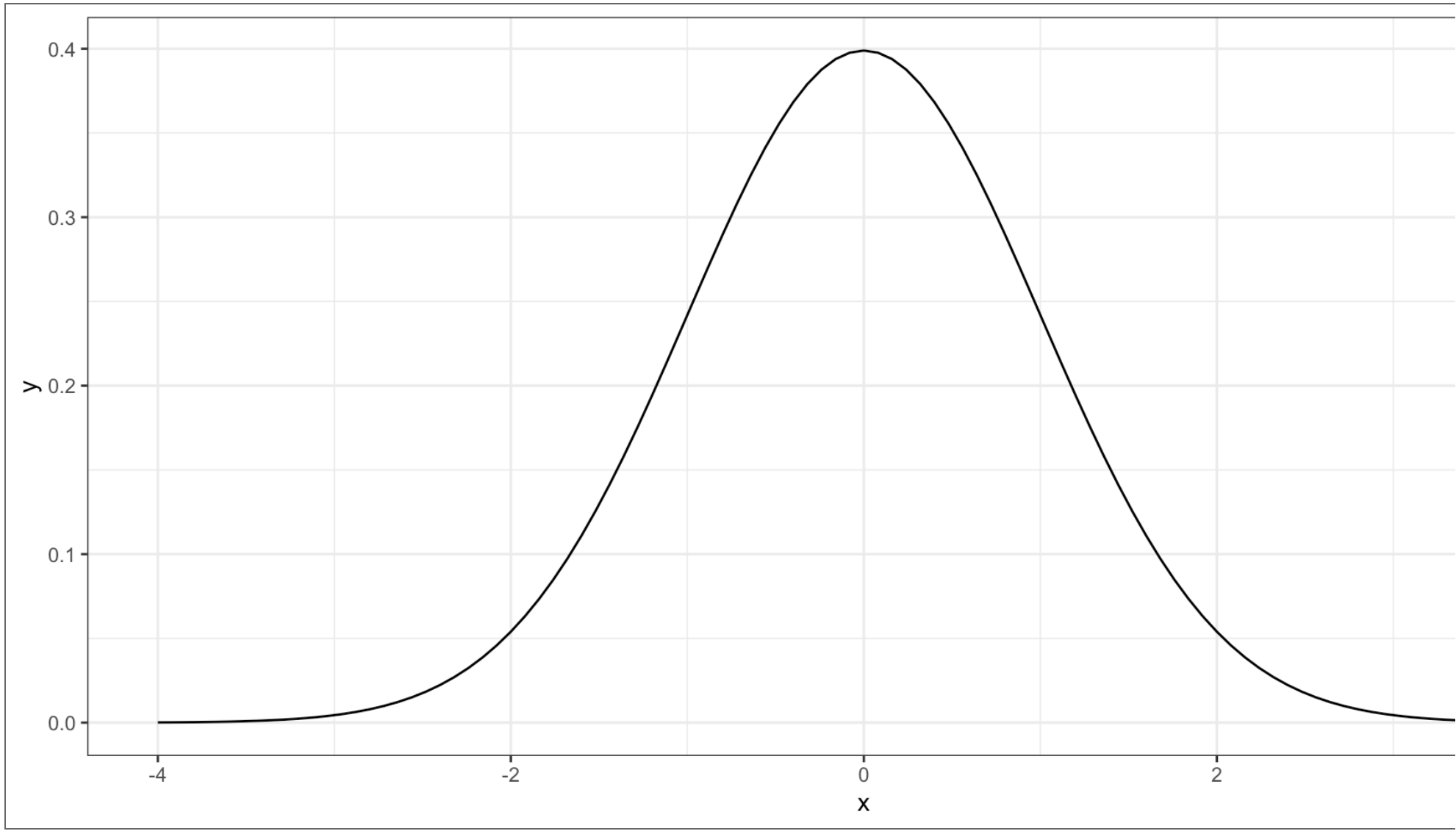
The Normal CDF



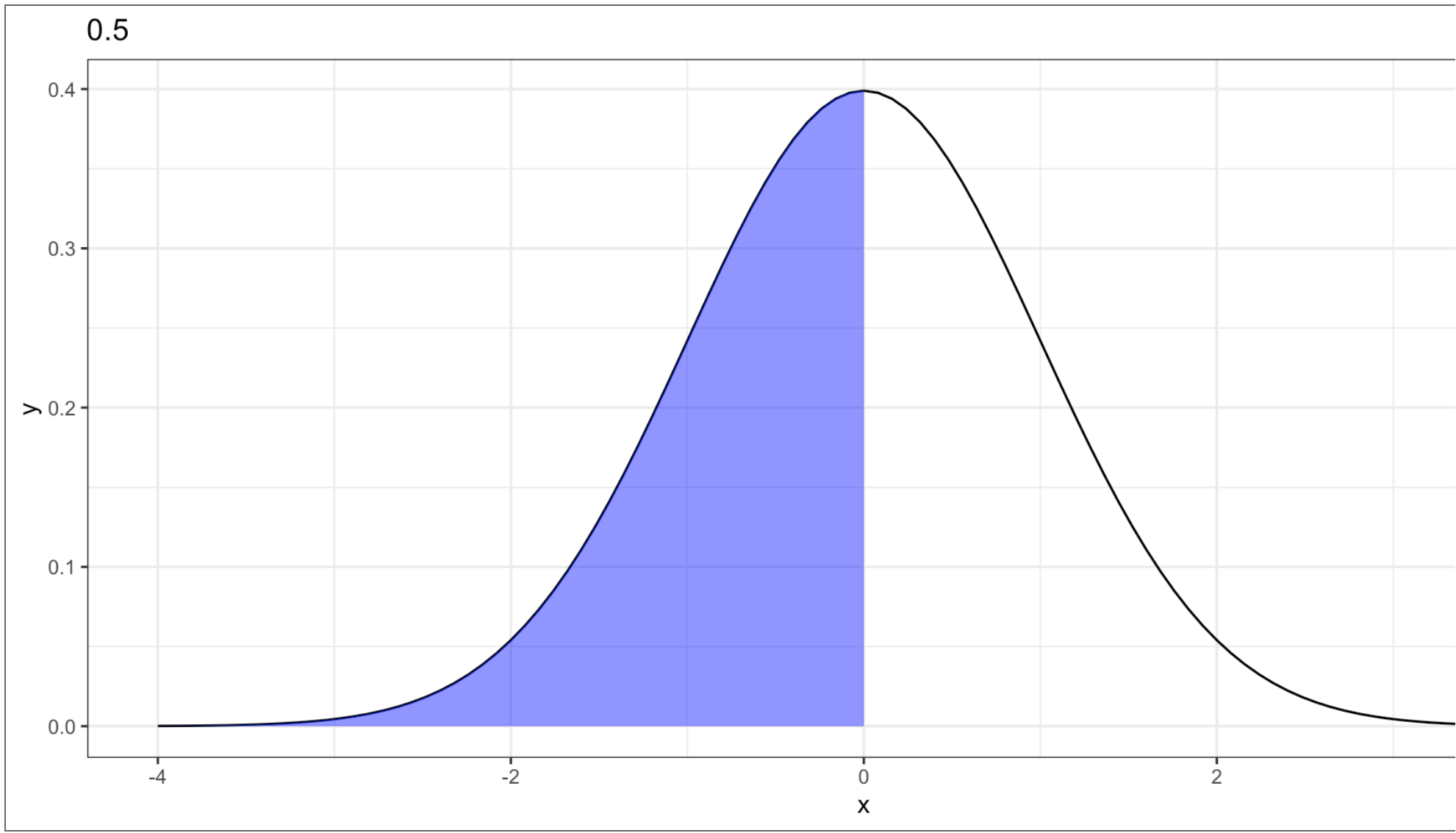
Special features of Normal distributions:

- The sum of many random variables from other distributions are often Normal
- For $X \sim N(\mu, \sigma^2)$, $Z = X + c$ is also Normal: $Z \sim (\mu + c, \sigma^2)$
- $Z = cX$ is distributed $Z \sim N(c\mu, (c\sigma)^2)$
- Z-scores of a Normal random variable are $N(0, 1)$

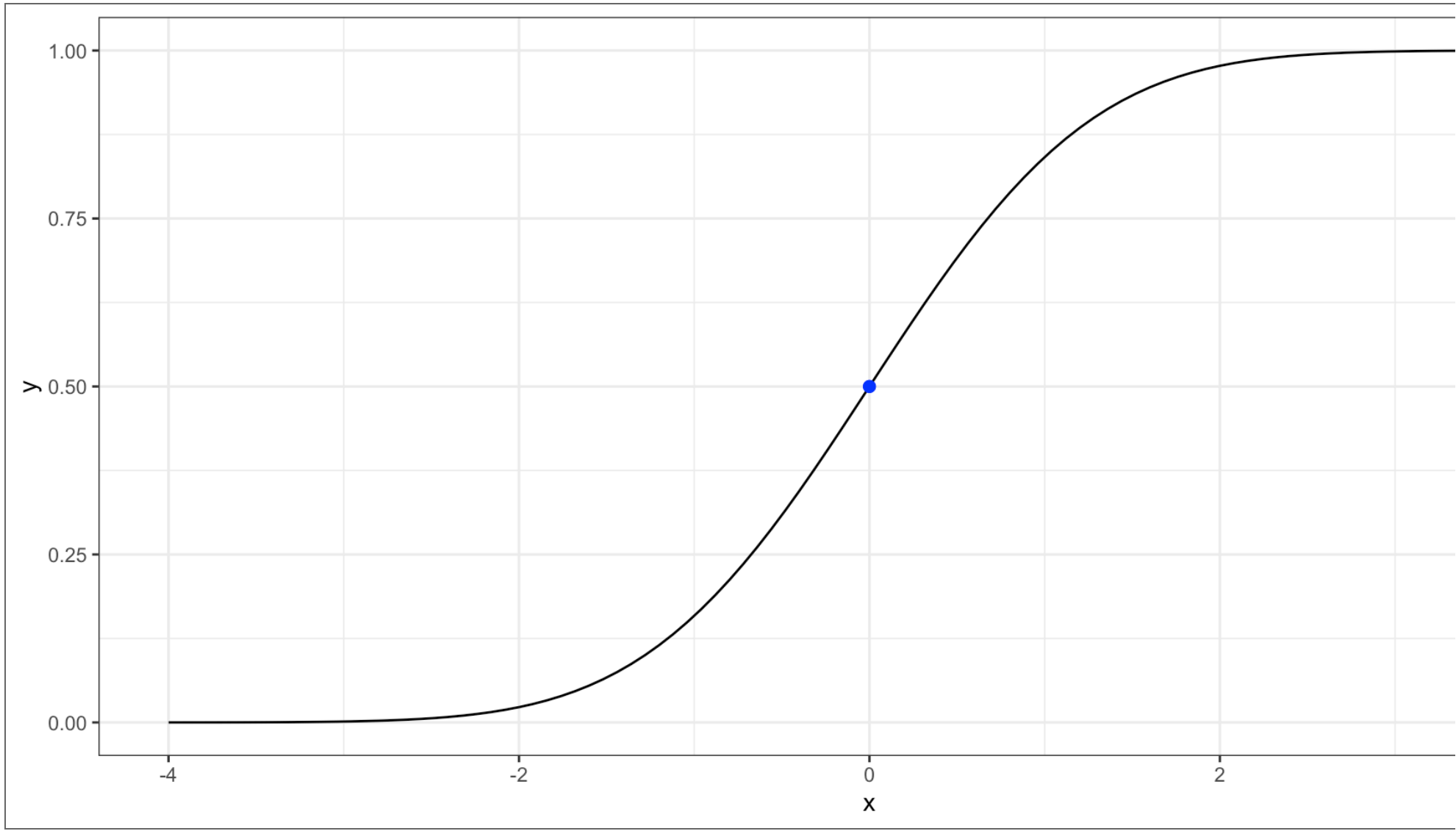
Area under the curve: interpreting the PDF and CDF



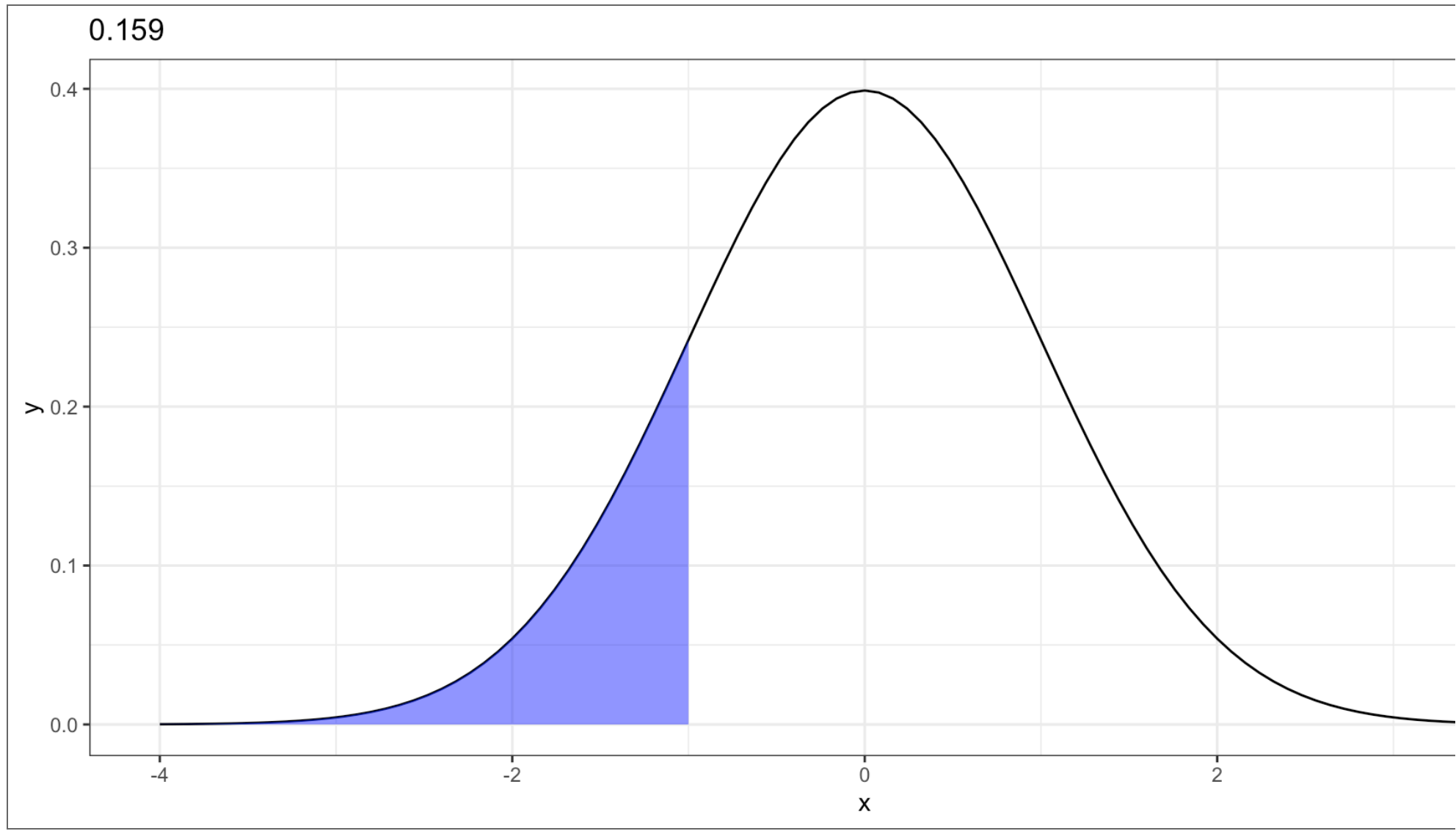
Area under the curve: interpreting the PDF and CDF



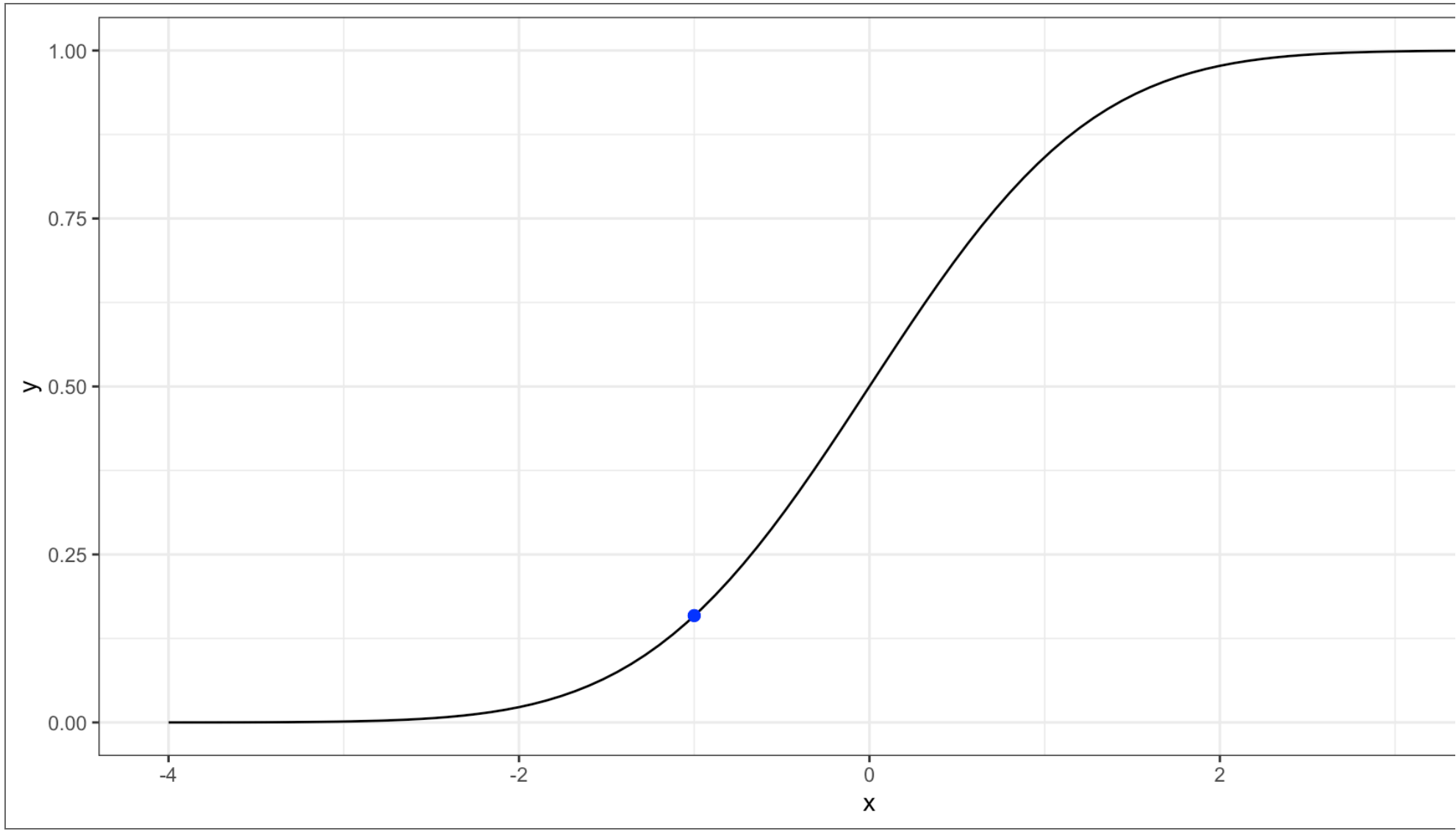
Area under the curve: interpreting the PDF and CDF



Area under the curve: interpreting the PDF and CDF



Area under the curve: interpreting the PDF and CDF



Z-scores and area under the curve

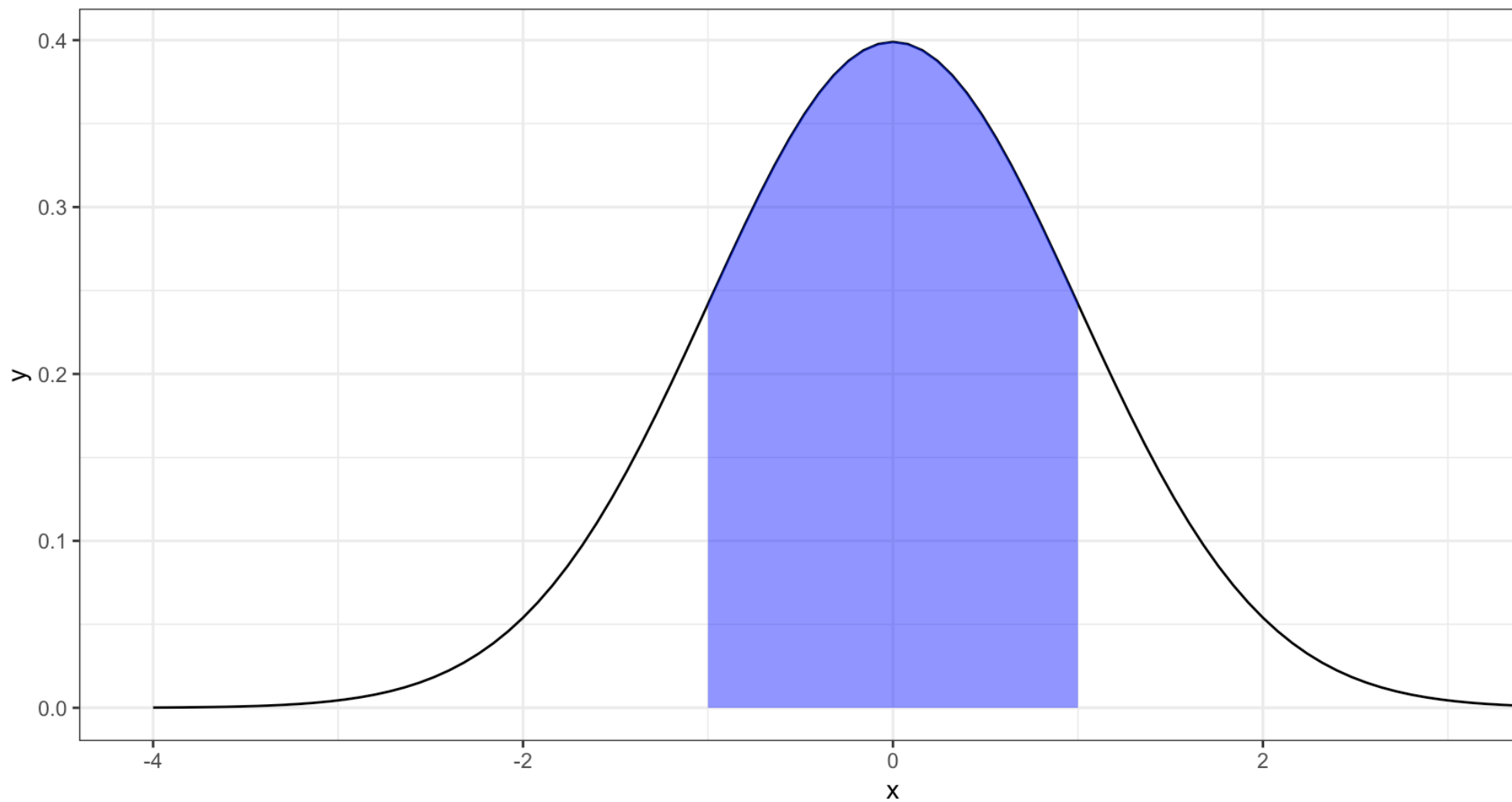
To obtain a z-score, we subtract the mean and divide by the standard deviation:

$$\text{z-score} = \frac{X - \mu}{\sigma}$$

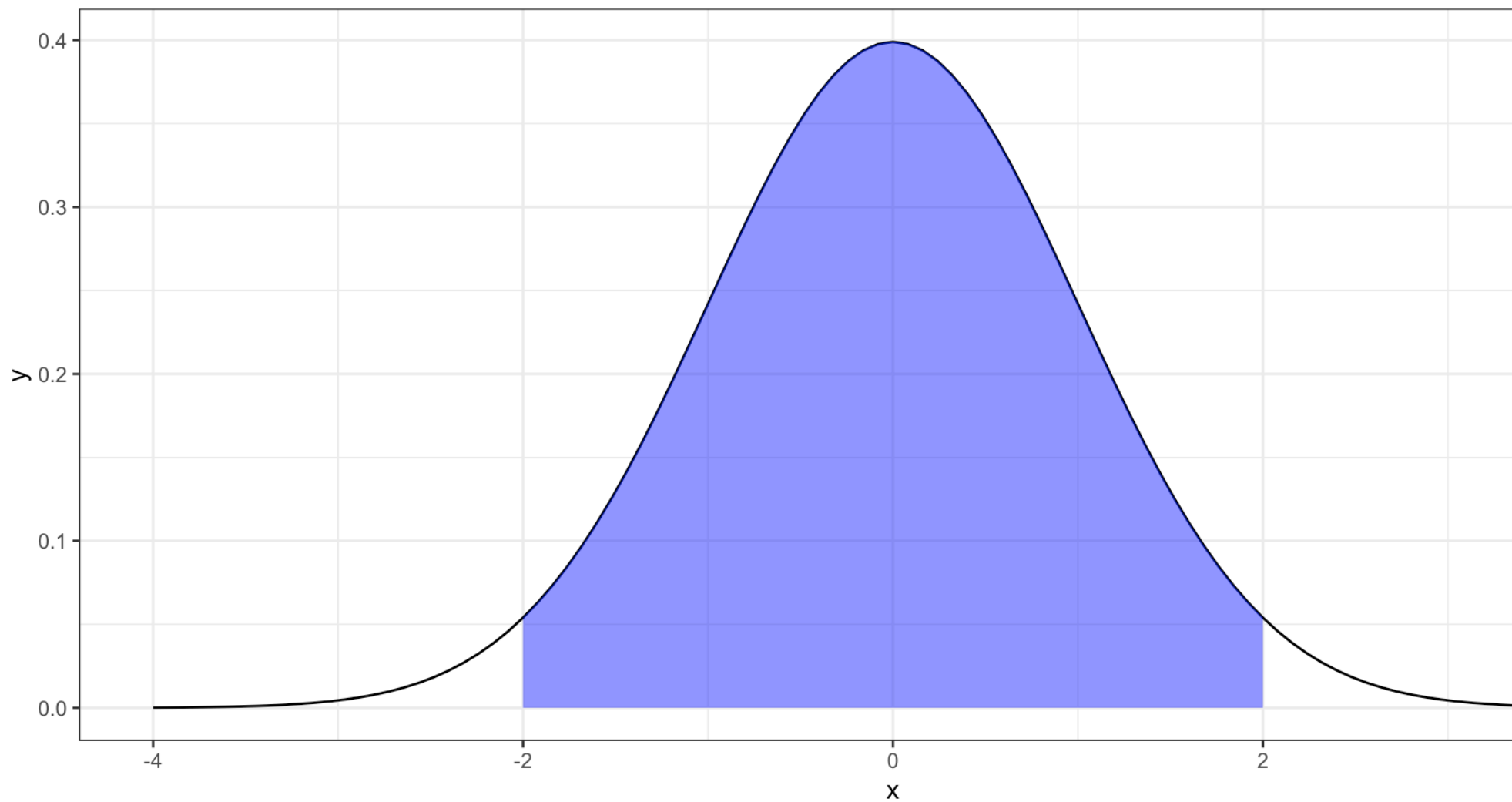
For a Normal variable, z-scores are distributed $z \sim N(0, 1)$

What does a z-score of 0 indicate? -1? 2?

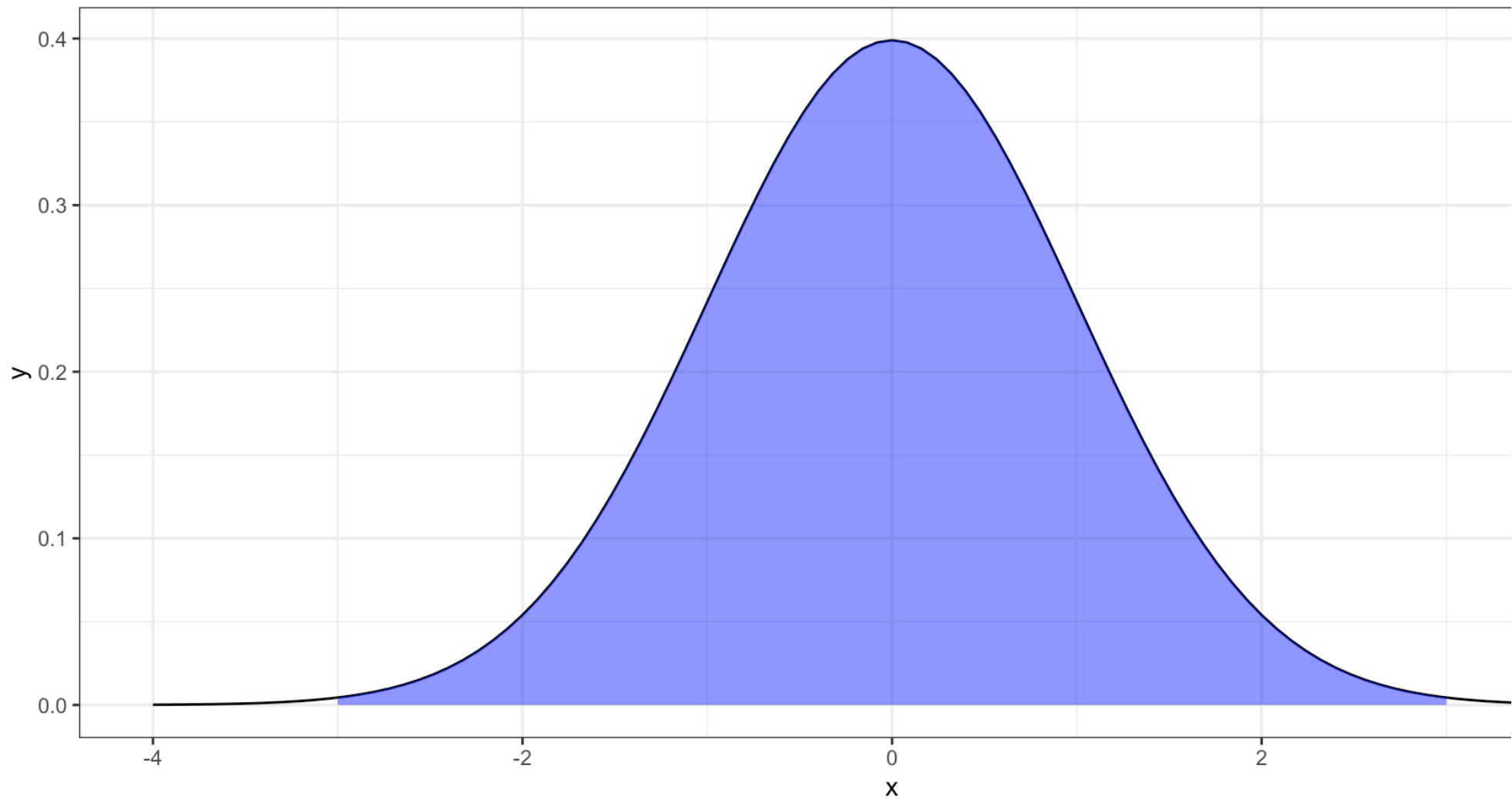
Mean $\pm$ 1 SD = 0.683



Mean +/- 2 SD = 0.954



Mean +/- 3 SD = 0.997



Useful probability distribution functions

```
1 ### Normal(0,1) probability density function
2 dnorm(x = 0, mean = 0, sd = 1)
```

```
[1] 0.3989423
```

```
1 ### Normal(0,1) cumulative distribution function
2 pnorm(q = 0, mean = 0, sd = 1)
```

```
[1] 0.5
```

```
1 ### Random draw from a normal(0,1) distribution
2 rnorm(n = 1, mean = 0, sd = 1)
```

```
[1] 0.7995289
```

```
1 ### CDF position for a given probability (quantile)
2 qnorm(p = 0.75, mean = 0, sd = 1)
```

```
[1] 0.6744898
```

1 ### You can also use `dbinom()`, `pbinom()`, `rbinom()`, `qbinom()`

Speaker notes